Code No: 117CD **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, December-2023/January-2024
DATA WAREHOUSING AND DATA MINING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions

PART – A

(25 Marks)

1.a)	What is fact table?	[2]
b)	What are different types of OLAP servers?	[3]
c)	What is data mining?	[2]
d)	What is the need of data preprocessing?	[3]
e)	What is closed frequent item set?	[2]
f)	What are the disadvantages of apriori algorithm?	[3]
g)	What is gini attribute selection measure?	[2]
h)	Distinguish between lazy learner and eager learner.	[3]
i)	What are various issues in k-means clustering algorithm?	[2]
j)	What are the various distance measures used in clustering algorithms?	[3]

PART – B

(50 Marks)

2. What is data warehouse? Explain data warehouse architecture in detail. [10]

OR

3. Draw star, snowflake and fact constellation schema for hospital management system. [10]

4. What are data mining tasks? Explain them in detail. [10]

OR

5. What is data transformation? Explain various data transformation techniques. [10]

6. Explain how FP growth algorithm overcomes the limitations of apriori algorithm. [10]

OR

7.a) Explain about confidence and support measures.
b) What is apriori property? Explain join and prune step in apriori algorithm. [5+5]

8. Write down algorithm for decision tree construction and explain various attribute selection measures. [10]

OR

9.a) Explain K-nearest neighbor classification.
b) What are various methods to evaluate a classifier? [5+5]

10. Explain hierarchical clustering algorithms in detail. [10]

OR

11. What is an outlier? Why outlier analysis is important? Explain various outlier detection methods. [10]

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